

AI DECLARATION FORMS

I used ChatGPT and Claude AI

Percentage : 20%

Below are a few screenshots of my interaction with the AI tools:

i did oop in java so when you are explaining this python version to me try to link it to java if possible

Perfect — that actually makes this *much easier*. I'll map everything directly to **Java OOP concepts** so you don't get confused.

1. Python `class GridProblem(Problem)` vs Java

Python:

`</> Python`



`▶ Run`

```
class GridProblem(Problem):
```

Java equivalent:

`</> Java`



```
class GridProblem extends Problem
```



 Same idea:

`+ s`



ChatGPT can make mistakes. Check important info.

Give me your own implementation of the best first search, I want to compare it to what I have please

Here's a clean implementation of **Best-First Search** based directly on the pseudocode in your lab. It uses:

- a priority queue (`heapq`)
- a **dictionary** for `reached`
- the **late goal test**
- replacement when a cheaper path to a state is found

 Python



 Run

```
import heapq

class BestFirstSearch(SearchAlgorithm):
    """Generic Best-First Search. Subclasses define evaluation()."""

    algorithm_name = "BestFirst"

    def __init__(self, heuristic=zero_heuristic):
        self.heuristic = heuristic

    def h(self, node: Node, problem: Problem) -> float:
        return self.heuristic(node.state, problem.goal)

    def evaluation(self, node: Node, problem: Problem) -> float:
        raise NotImplementedError(
            "Subclasses must implement evaluation(node, problem)"
        )
```

Now we've found the real issue.

Your `BestFirstSearch` is using a `PriorityQueue` class that behaves differently from `heapq`, and your `SearchResult` field names are also inconsistent with your earlier lab version.

But the **current error you were getting is still the same root problem:**

✖ Node objects are being compared inside the priority queue

🚨 MAIN PROBLEM IN YOUR CODE

This line is dangerous:

```
</> Python
frontier.push(self.evaluation(node, problem), node)
```

and:

```
</> Python
frontier.push(self.evaluation(child, problem), child)
```

Why?

+ Ask anything

```
TypeError                                Traceback (most recent call last)
Cell In[37], line 25
    21 watar2 = WeightedAStarSearch(manhattan_distance, weight=2)
    22 ucs = UniformCostSearch()
    23
    24 uniform_results = [
--> 25     ucs.search(uniform_problem),
    26     astar.search(uniform_problem),
    27     watar2.search(uniform_problem),
    28     greedy.search(uniform_problem),
```

Show more ▾

Good — this traceback tells us something very specific:

- 👉 The bug is NOT in `ucs.search(...)` itself
- 👉 It is inside your `BestFirstSearch.search()`, but you are still running an **old version**

```
uniform_results = [

    astar.search(uniform_problem),
    watar2.search(uniform_problem),
    greedy.search(uniform_problem),
].....the prob;em is still coming from here
```

Good — now we're actually at the real issue. The error is NOT in that list.

This line:

<> Python

```
uniform_results = [
    astar.search(uniform_problem),
    watar2.search(uniform_problem),
    greedy.search(uniform_problem),
]
```



Run



yes, the error is from here

```
uniform_results = [
    ucs.search(uniform_problem),
    astar.search(uniform_problem),
    watar2.search(uniform_problem),
    greedy.search(uniform_problem),
]

show_results(uniform_results)
```

1:13 PM ↺ ✎ 📄

I'd like to see the actual traceback to pin this down precisely — could you paste it? In the meantime, here are the most likely culprits given everything we've built so far, roughly in order of probability:

1. `UniformCostSearch` / `GreedyBestFirstSearch` / `AStarSearch` not implemented yet (or only `BestFirstSearch.search` is)



If those subclasses only define `evaluation()` and rely on inheriting `search()` from

Write a message...

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